

Adverse Selection I - part 2

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Readings: MSPC Chapter 4.1-4.2

Timing

- We now look at a more formal problem
- Nature chooses a type of agent A
 - type G with probability q or B with probability $(1-q)$
- P designs and offers a contract
- A either accepts or rejects
- A supplies *verifiable* effort
- Nature plays (draws random profit levels depending on effort)
- Outcomes and payoffs are realised

Note that the incomplete information is now about the type of the agent and *not* about effort. Problem arises because only A knows his type and thus can pick any contract.

Utility functions

- Principal: $\Pi(e) - w$
 - We model gross profits as $\Pi(e)$, with $\Pi'(\cdot) > 0$ and $\Pi''(\cdot) < 0$.
 - Note that effort is verifiable here and P is risk neutral in money.
 - The assumptions ensure a concave objective function.
- Agent: Type G : $u(w) - v(e)$
- Agent: Type B : $u(w) - kv(e)$, where $k > 1$.
 - Assume $u'(\cdot) > 0$ and $u''(\cdot) < 0$, $v'(\cdot) > 0$, $v''(\cdot) > 0$, and $v(0) = 0$.
 - Reservation utility $\bar{u}$.

In the first best, P can observe A 's type.

- The problem of the principal contracting with a type G agent can be written as:

$$\max_{[e,w]} \Pi(e) - w$$

subject to

$$u(w) - v(e) \geq \underline{u}$$

- and similarly for a type B agent:

$$\max_{[e,w]} \Pi(e) - w$$

subject to

$$u(w) - kv(e) \geq \underline{u}$$

Optimality conditions

- IR will be binding.
- For a type G agent:

$$u(w_G^*) - v(e_G^*) = \underline{u}$$

$$\Pi'(e_G^*) = \frac{v'(e_G^*)}{u'(w_G^*)}$$

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- Graphical illustration in Figure 4.3 in the book.
- Generally, as can be seen in the graph, $e_B^* < e_G^*$, but it is not clear whether $w_B^* \geq w_G^*$, since a type B requires larger compensation for a given level of effort (i.e. w_B^* is higher for a given level of effort) but also is required to provide less effort.

Second best - Asymmetric information

- P does not observe the type of agent he is dealing with.
- Now the contracts (e_G^*, w_G^*) and (e_B^*, w_B^*) are not a good deal for the principal since:

$$u^G(w_B^*, e_B^*) = u(w_B^*) - v(e_B^*) > u(w_B^*) - kv(e_B^*) = \underline{u}$$

- The LHS is utility for type G under contract (w_B, e_B) , which must give strictly more than contract (w_G, e_G) . Thus type G agents will pretend to be type B .

Revelation principle

- In principle, the principal could offer anything as a contract.
- The revelation principle states that in the search for the optimal contract, the principal can restrict her search for the optimal contract to the set of truthful direct revelation mechanisms, that are of the following form:
 - The agent must announce his type, after he observes it.
 - The contract specifies a payment to the agent for each possible announcement (i.e. for each possible type).
 - The contract is designed so that the agent has incentive to reveal his type truthfully.

Contract choices

- Using the revelation principle we can focus on the set of truthful direct revelation mechanisms.
- This requires that we ensure that contracts are self-selective or incentive compatible
- The principal must offer an incentive compatible menu of wage-effort pairs.
- The principal can offer a **pooling** contract or a **separating** contract.
- Here we focus on the latter case.

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$$u(w_G) - v(e_G) \geq u(w_B) - v(e_B) \quad (IC_G)$$

$$u(w_B) - kv(e_B) \geq u(w_G) - kv(e_G) \quad (IC_B)$$

It is often the case that the number of constraints can be reduced by identifying non-binding constraints and removing them from the problem. This makes the problem easier to solve. In the problem above note that:

$$u(w_G) - v(e_G) \geq u(w_B) - v(e_B) > u(w_B) - kv(e_B) \geq \underline{u}$$

This implies that the participation constraint for type G is redundant and can be removed from the principals problem above.

Two cases

We can also combine the two IC constraints:

$$v(e_G) - v(e_B) \leq u(w_G) - u(w_B) \leq k[v(e_G) - v(e_B)]$$

Since $k > 1$ and LHS and RHS are otherwise identical this implies $v(e_G) \geq v(e_B)$ and thus $e_G \geq e_B$. Thus greater (or equal) effort will be demanded from the most efficient agent also under incomplete information.

Comments

Two cases stand out:

- Case 1: if $e_G = e_B$ the combined constraints also says that $w_G = w_B$. This seems unlikely to be optimal: since type G 's IR constraint is not binding it is plausible that P could increase e_G and w_G slightly and attract only type G .
- Case 2: if $e_G > e_B$, then both IC constraints cannot bind simultaneously, since if:

$$u(w_G) - v(e_G) = u(w_B) - v(e_B)$$

then

$$u(w_G) - kv(e_G) < u(w_B) - kv(e_B)$$

$$\frac{\partial \mathcal{L}(\cdot)}{\partial w_G} = -q + \mu u'(w_G) - \delta u'(w_G) = 0 \quad (1)$$

$$\frac{\partial \mathcal{L}(\cdot)}{\partial w_B} = -(1 - q) + \lambda u'(w_B) - \mu u'(w_B) + \delta u'(w_B) = 0 \quad (2)$$

$$\frac{\partial \mathcal{L}(\cdot)}{\partial e_G} = q\Pi'(e_G) - \mu v'(e_G) + \delta kv'(e_G) = 0 \quad (3)$$

$$\frac{\partial \mathcal{L}(\cdot)}{\partial e_B} = (1 - q)\Pi'(e_B) - \lambda kv'(e_B) + \mu v'(e_B) - \delta kv'(e_B) = 0 \quad (4)$$

- Arrive at:

$$\lambda = \frac{q}{u'(w_G)} + \frac{1-q}{u'(w_B)} > 0 \quad (5)$$

- From this we see that (PC_B) must be binding (the Lagrange multiplier is positive).
- Isolate for $\mu - \delta k$ in (3) and insert in (4):

$$\lambda = \frac{q\Pi'(e_G)}{kv'(e_G)} + \frac{(1-q)\Pi'(e_B)}{kv'(e_B)} \quad (6)$$

Summary

- We have shown that $\mu > 0$ and $\lambda > 0$.
- Consider two cases:
 - Case 1: $\delta > 0$ (this is also when $e_G = e_B = e$ and $w_G = w_B = w$ as argued above)
 - Case 2: $\delta = 0$

We can now go through the two cases.

- Case 1: $e_G = e_B = e$ and $w_G = w_B = w$
- In this case the two expressions for λ (5) and (6) reduce to:

$$\frac{kv'(e)}{u'(w)} = \Pi'(e)$$

- Compared to the first best, effort is now too low for type G , and as argued above the principal will have an incentive to try to increase effort for G by offering slightly higher wages and effort.
(see the book for a formalised mathematical argument)

- Case 2: $e_G > e_B$: $\delta = 0$
- In this case, (1) and (3) imply:

$$\frac{v'(e_G)}{u'(w_G)} = \Pi'(e_G) \quad (7)$$

which is the same as the efficiency condition in the first best case (marginal effort condition not distorted)

- this is also known as "no distortion at the top" or more precisely; *the contract designed for the agent no one wants to mimic is not distorted.*

Thus, we have:

- Unknowns: e_G, e_B, w_G, w_B
- Binding (IC_G)
- Binding (PC_B)
- Using (1) and (3) we have

$$\frac{v'(e_G)}{u'(w_G)} = \Pi'(e_G)$$

- we need one more equation.

Expressions (5) and (6) reduce to a relation which defines the effort level for type B :

$$\begin{aligned}
 & \frac{q \frac{v'(e_G)}{u'(w_G)}}{kv'(e_G)} + \frac{(1-q)\Pi'(e_B)}{kv'(e_B)} = \frac{q}{u'(w_G)} + \frac{1-q}{u'(w_B)} \\
 \Leftrightarrow & \frac{qv'(e_B)}{(1-q)u'(w_G)} + \Pi'(e^B) = \frac{qkv'(e_B)}{(1-q)u'(w_G)} + \frac{kv'(e_B)}{u'(w_B)} \\
 \Leftrightarrow & \Pi'(e^B) = \frac{qv'(e_B)(k-1)}{(1-q)u'(w_G)} + \frac{kv'(e_B)}{u'(w_B)} \quad (8)
 \end{aligned}$$

This RHS is increasing in q . This means that the more type G agents there are the lower effort the principal demands from type B (the more profitable is it for him to distort the type B contract instead of paying more to type G in order to make sure good types do not mimic bad types).

- Summing up, the contract under incomplete information is defined by 4 equations: (7), (8), (PC_B) and (IC_G) .
- $e_G > e_B$ and $w_G > w_B$.
- Type G gets informational rent. This can be seen from the binding (IC_G) and (PC_B) :

$$\begin{aligned}u(w_G) - v(e_G) &= u(w_B) - v(e_B) \\&= u(w_B) - kv(e_B) + (k-1)v(e_B) \\&= \underline{u} + (k-1)v(e_B)\end{aligned}$$

Summary

- Type G is offered an efficient contract. This is known as no distortion at the top or more precisely; the contract designed for the agent no-one wants to mimic is not distorted
 - NB: this does not necessarily imply that the first best and second best levels coincide
 - if the agent is risk-neutral, (7) does not depend on w_G and $e_G^{FB} = e_G^{SB}$.
 - if the agent is risk-averse, (7) implies that $e_G^{FB} \neq e_G^{SB}$ since $w_G^{FB} \neq w_G^{SB}$ (due to rent). The difference in wage payments compared to the first best “spills over” into the effort requirement. Nevertheless for a given wage the required level of effort is efficient.

- Type B 's contract is distorted compared to the first best (for a given wage, under perfect information P would demand higher effort from type B).
- This happens to ensure that type G does not pretend to be type B .
- By distorting the contract the principal loses some efficiency and produces sub-optimally for type B (compared to under complete information) **but** it allows P to reduce the informational rent which depends on e_B .

- A final possibility exists.
- The principal could only offer the first best type- G contract
- and offer no contract that would be attractive to the type- B agents.
- Optimality of this requires

$$q[\Pi(e_G^{SB}) - w_G^{SB}] + (1 - q)[\Pi(e_B^{SB}) - w_B^{SB}] \leq q[\Pi(e_G^{FB}) - w_G^{FB}]$$